

Om Chavda

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EDUCATION

Dhirubhai Ambani University

B.Tech in Information and Communication Technology, CGPA: 8.64/10

Gandhinagar, Gujarat

Aug. 2023 – Present

B.N. Virani Higher Secondary School

High School (Science), Result: 99.21 percentile

Bhavnagar, Gujarat

Jun. 2021 – Jun. 2023

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, C++, C, SQL, Python, Rust, HTML5, CSS

Frameworks/Technologies: Next.js, Express.js, React.js, Node.js, Socket.IO, Tailwind CSS, Redux Toolkit

Databases & ORMs: PostgreSQL, Redis, SQLite, pgvector, MongoDB, Prisma ORM, Drizzle ORM

DevOps & Cloud: Docker, Nginx, AWS EC2, Cloudflare, GitHub Actions (CI/CD), Vercel, Render

Developer Tools: VS Code, Git, GitHub, Linux, Postman

Core CS: OOP, DBMS, Operating Systems, Computer Networks, Low Level Design, High Level Design

PROJECTS

Dokit – Cloud-Native Collaborative Development Workspace

[Live](#) | [GitHub](#)

Next.js, Express, TypeScript, PostgreSQL, Redis, Docker, Socket.IO, Yjs, Nginx, AWS EC2, Cloudflare R2

- Architected on-demand provisioning and teardown of isolated, non-root Docker sandboxes via the Docker Engine API, giving each workspace a secure containerized runtime with full terminal access.
- Developed conflict-free, real-time collaborative code editing with Yjs CRDTs and CodeMirror 6, supporting live cursors, presence tracking, and multi-user simultaneous editing.
- Integrated a project-aware AI assistant (ASTra) with a full RAG pipeline – regex-based, multi-language AST chunking, local Ollama embeddings, pgvector similarity search with MMR reranking, and persistent chat history.
- Designed bidirectional file synchronization between containers and the browser using Linux inotify watchers and Socket.IO, keeping the editor's file tree instantly consistent with runtime changes.
- Engineered wildcard Nginx routing (`[port]-[projectId].dokit.backends.live`) to expose live dev-server previews instantly, with authenticated proxying for both terminal WebSocket sessions and preview HTTP traffic.
- Persisted workspace code to Cloudflare R2 via BullMQ background jobs, enabling restoration on project restart.
- Developed secure code-snippet sharing (Code Links) with Argon2-encrypted password protection, configurable visibility – public, anyone with a link, or restricted to specific allowed users – and link expiration.
- Configured short-lived JWT access tokens with rotating refresh tokens, remote session revocation, TOTP-based 2FA and backup recovery codes, GeoIP session tracking, and Redis sliding-window rate limiting.
- Deployed on AWS EC2 with a 3-version rolling deployment strategy – atomic symlink swap per release with health-check-gated automatic rollback.

Ingenium – Autonomous Local AI Coding Agent

[GitHub](#)

TypeScript, Node.js, OpenRouter, Telegram Bot API, SQLite, Drizzle ORM, Tavily

- Programmed a local agentic CLI with Agent Mode (autonomous execution with staged approval) and Plan Mode (step-by-step plan written to `ingenium-plan.md`, reviewable before execution).
- Established a Telegram Bot interface for remote execution, session switching, inline file diffs, and approval callbacks from any device without physical machine access.
- Structured a modular tool-use architecture covering file system, git, shell, process management, and web search via Tavily, with persistent cross-session memory backed by SQLite and Drizzle ORM.
- Added a background daemon for natural-language reminder scheduling and a transient session mode where no conversation history is persisted to disk.

ACHIEVEMENTS

- Solved **2100+** problems across LeetCode (**chavdaom84**) and Codeforces (**Om_007**).
- **Knight** on LeetCode (peak rating **2073**); **Specialist** on Codeforces (peak rating **1451**).
- Secured global rank **427** in ICPC 2025 Amritapuri Regional Preliminary Round.
- Achieved rank **580** out of 39,000+ participants in LeetCode Biweekly Contest 186.
- Placed **694** out of 38,000+ participants in LeetCode Weekly Contest 508.
- **3rd Prize** in Winter of Code 7.0 at MSTC, Dhirubhai Ambani University among 50+ participants.